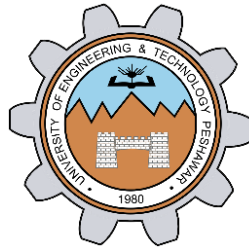


LAB # 02

**INTRODUCTION TO TOP-DOWN APPROACH AND DATA
FLOW LEVEL MODELING**



Spring 2024

CSE-308L Digital System Design Lab

Submitted by: **Shahzad Bangash**

Registration No: **21PWCSE1980**

Class Section: **C**

“On my honor, as student at University of Engineering and Technology, I have
neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Engr. Shahzada Fahim Jan

February 25, 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

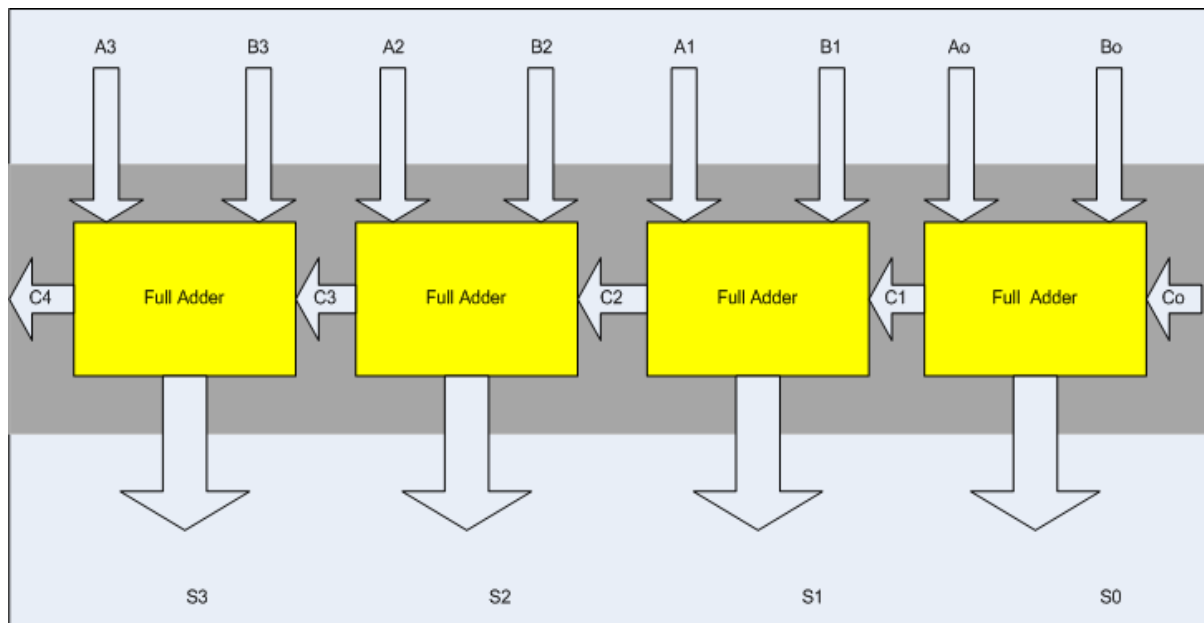
Objectives of The Lab:

- Learn top down and bottom-up design methodologies
- Data flow level modeling

Lab Tasks:

Task 1a:

4-bit Ripple Carry Adder



- 1- First implement a Full adder using data gate level modeling.
- 2- Simulate the Full adder with a test bench.
- 3- Instantiate the Full adder four times and connect the circuit as shown.
- 4- Now again write a test bench and simulate the 4 bit RCA.

1. Full Adder:

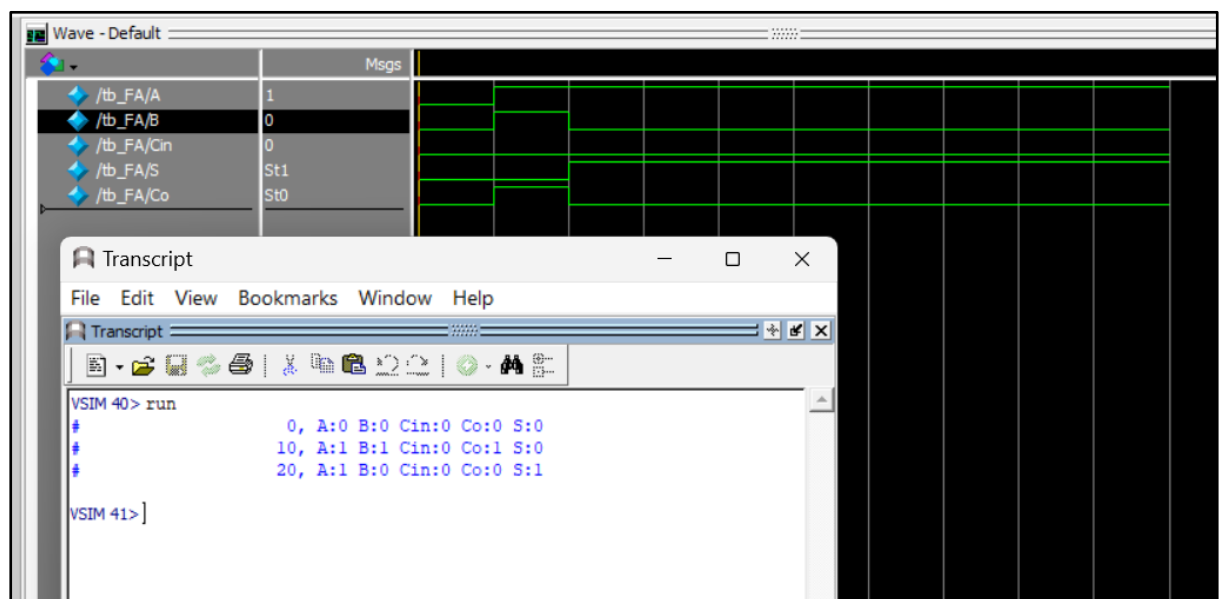
DUT:

```
1  module FA(S,Co,A,B,Cin) ;
2      input  A,B,Cin;
3      output S,Co;
4      wire  y1,y2,y3,y4;
5      //S = Cin XOR (A XOR B)
6      xor x1(y1,A,B) ;
7      xor x2(S,y1,Cin) ;
8
9      //Co = AB + Cin.(A XOR B)
10     xor x3(y2,A,B) ;
11     and a1(y3,y2,Cin) ;
12     and a2(y4,A,B) ;
13     or o1(Co,y3,y4) ;
14 endmodule
```

2. FA Test Bench:

```
1  module tb_FA();
2      reg A,B,Cin;
3      wire S,Co;
4
5      FA f1(S,Co,A,B,Cin);
6  initial begin
7      A=0; B=0; Cin=0;
8      #10; A=1; B=1; Cin=0;
9      #10; A=1; B=0; Cin=0;
10 end
11     initial
12         $monitor("%d, A:%b B:%b Cin:%b Co:%b S:%b", $time, A, B, Cin, Co, S);
13 endmodule
```

Output:



3. RCA DUT

```
1  module Ripple_Carry_Adder(S,Co,A,B,Cin);
2      input [3:0]A,B;
3      input Cin;
4      output [3:0]S;
5      output Co;
6      wire c1,c2,c3;
7      FA fa1(S[0],c1,A[0],B[0],Cin);
8      FA fa2(S[1],c2,A[1],B[1],c1);
9      FA fa3(S[2],c3,A[2],B[2],c2);
10     FA fa4(S[3],Co,A[3],B[3],c3);
11 endmodule
```

4. 4-bit RCA Test Bench

```
1 module tb_Ripple_Carry_Adder();
2 reg [3:0] A,B;
3 reg Cin;
4 wire [3:0] S;
5 wire Co;
6
7 Ripple_Carry_Adder RCA1(S,Co,A,B,Cin);
8
9
10 initial begin
11     A=4'b0000; B=4'b0000; Cin=0;
12     #10; A=4'b1001; B=4'b1001; Cin=0;
13     #10; A=4'b1011; B=4'b1011; Cin=0;
14
15 end
16
17 //initial
18 // $monitor("S, Co, A, B, Cin",S,Co,A,B,Cin);
19 initial
20     $monitor("%d, S:%b, Co:%b, A:%b, B:%b, Cin:%b", $time, S, Co, A, B, Cin);
21 endmodule
```

Output:

The screenshot displays a simulation environment with two main windows. The top window, titled 'Wave - Default', shows a list of signals on the left: /tb_Ripple_Carry_Adder/A (1011), /tb_Ripple_Carry_Adder/B (1011), /tb_Ripple_Carry_Adder/Cin (0), /tb_Ripple_Carry_Adder/S (0110), and /tb_Ripple_Carry_Adder/Co (St1). The main area of this window shows a timing diagram with three columns of data: 0000, 1001, and 1011. The bottom window, titled 'Transcript', contains a menu bar (File, Edit, View, Bookmarks, Window, Help) and a toolbar. The transcript text is as follows:

```
add wave sim:/tb_Ripple_Carry_Adder/*
VSIM 44> run
#           0, S:0000, Co:0, A:0000, B:0000, Cin:0
#           10, S:0010, Co:1, A:1001, B:1001, Cin:0
#           20, S:0110, Co:1, A:1011, B:1011, Cin:0
VSIM 45> ]
```

Task 1b:

Design the 4 bit full adder using data flow level modeling.

DUT:

```
1  module Full_Adder_Data_Flow(S,Co,A,B,Cin);  
2  input A,B,Cin;  
3  output S,Co;  
4  
5  assign S = A^B^Cin;  
6  assign Co = A&B | Cin&(A^B);  
7  
8  endmodule
```